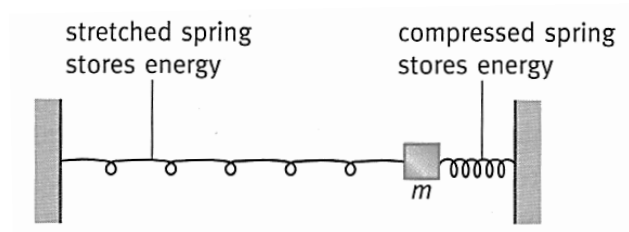


13. An object of mass m oscillates in simple harmonic motion between two springs as shown below. The amplitude of the motion is x_0 .



What is the ratio $\frac{\text{elastic potential energy of the two springs}}{\text{kinetic energy of the object}}$ when the object is $\frac{x_0}{2}$ from its equilibrium?

- A $\frac{1}{3}$ B $\frac{1}{2}$ C 1 D 3

Ans: A

Let the total energy of the oscillating system be $E = \frac{1}{2} m \omega^2 x_0^2$.

Since potential energy of the system is $\frac{1}{2} m \omega^2 x^2$, when $x = \frac{1}{2} x_0$, the potential energy is $\frac{E}{4}$. Then the kinetic energy would have to be $\frac{3E}{4}$. Hence ratio is $\frac{1}{3}$.

14 X and Y are two points on the surface of water in a ripple tank. A source of constant